Week 5 Workshop

Question 1
Download monthly data from 'Yahoo Finance' on Microsoft (MSFT) from December 2015 to December 2025. Consider the CAPM regression

$$r_t - r_{ft} = \alpha + \beta_1 \left(r_{mt} - r_{ft}\right) + \varepsilon_t$$

where r_t is the monthly log return on MSFT, $r_{mt} - r_{ft}$ is the market factor (MKT-RF), and r_{ft} is the risk-free rate (RF). Download the monthly market factor (MKT-RF) and risk-free rate (RF) from Ken French's website.

(a) Estimate the CAPM over the following sub-periods:

(i) January 2016 to December 2020

(ii) January 2021 to December 2025

Report the estimated beta and its standard error for each sub-period.

(b) Test whether MSFT tracks the market.

(c) For the first sub-period, report the idiosyncratic risk and explain what it tells you.

Question 2
Consider the multi-factor model:

$$r_t - r_{ft} = \alpha + \beta_1 \left(r_{mt} - r_{ft}\right) + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \varepsilon_t$$

where r_t is the monthly return on the Cnsmr industry portfolio, $r_{mt} - r_{ft}$ is the market factor (MKT-RF), r_{ft} is the risk-free rate (RF), SMB is the size factor, HML is the value factor, MOM is the momentum factor. Download the monthly data from Ken French's website. Estimate the model using data from January 1997 to December 2025. Test for the validity of the CAPM over the multi-factor model.